LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

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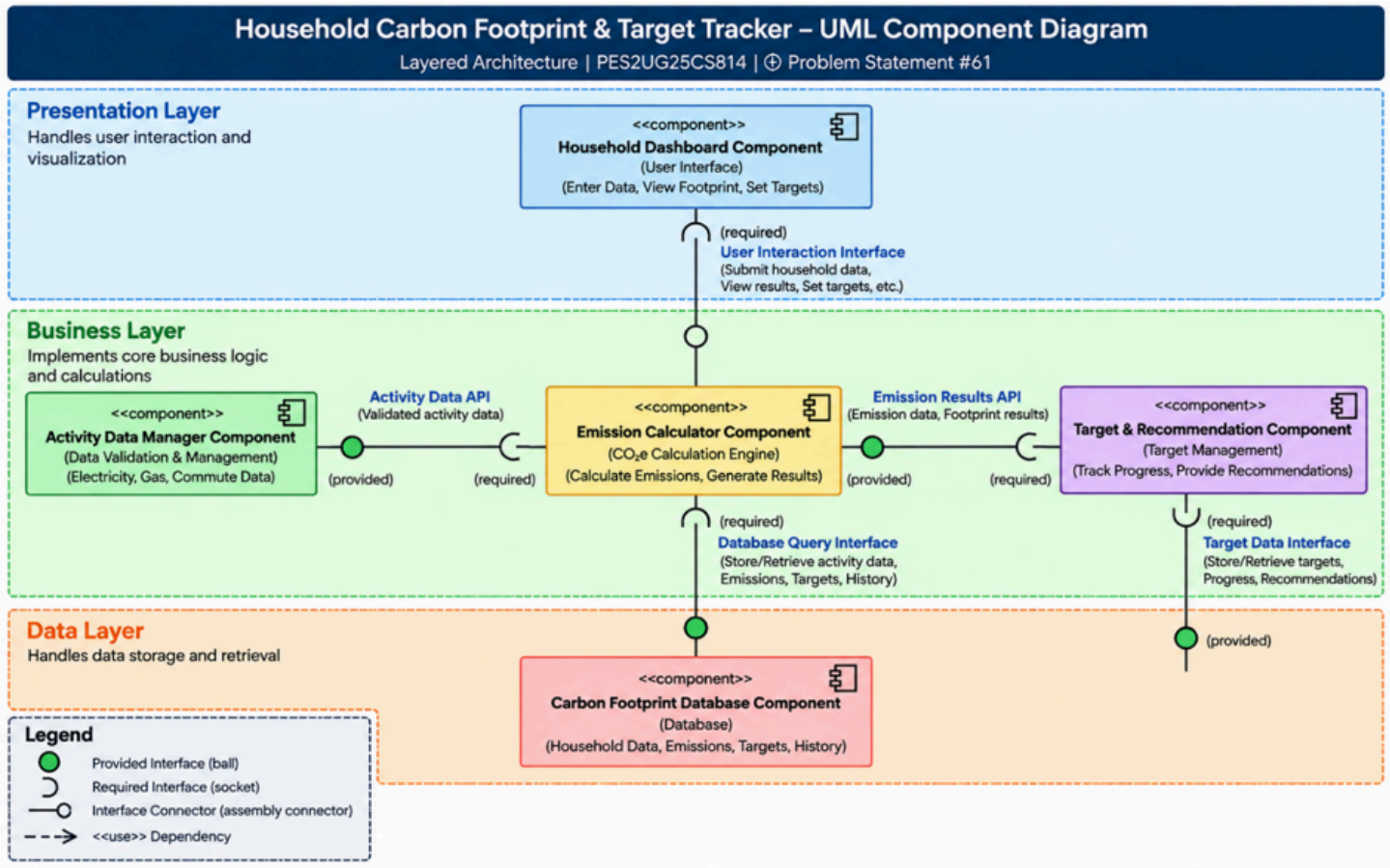
Title: Household Carbon Footprint & Target Tracker

Interface Description:

Interface	Purpose
Activity Data API	Sends household activity data from Dashboard to Data Manager
Emission Calculation API	Sends validated activity data to Emission Calculator
Emission Results API	Sends calculated emissions to Target & Recommendation
Carbon Data Query	Retrieves/stores carbon footprint information
Target & History API	Stores and retrieves targets and historical information

UML Component Diagram-Layered Architecture

HOUSEHOLD CARBON FOOTPRINT & TARGET TRACKER



Architecture Selection

We chose Layered Architecture for the Household Carbon Footprint & Target Tracker System.

Architectural Choice

The system uses Presentation, Business Logic, and Data layers. The Presentation layer provides the household dashboard, the Business Logic layer handles activity management, emission calculation, targets and recommendations, and the Data layer stores household carbon footprint information.

Reason 1

Layered Architecture provides clear separation of concerns. The dashboard is separated from the carbon calculation and data storage components, making the system easier to understand and maintain.

Reason 2

The architecture allows individual functions such as emission calculation, target tracking, and data management to be developed and modified separately without changing the entire system.

Security Advantage

Separating the presentation, business logic, and database layers reduces direct access to stored household information. Data validation and access control can be handled in the business logic layer before information reaches the database.

Performance Benefit

Separating the components allows frequently used operations such as viewing the carbon footprint to be handled efficiently without directly exposing the database to the user interface.

Conclusion

The Layered Architecture provides a clear structure for the Household Carbon Footprint & Target Tracker by separating user interaction, emission calculations, target tracking, and data storage. This makes the system easier to maintain, secure, and improve in the future.